Achernar

Minimum Necessary Data Collection

Public-data-first development / v6

Revision	October 12, 2026
Status	Research-backed plan; implementation and fieldwork pending
Architecture	`achernar_v6.tex`
Evidence	`docs/public-dataset-audit-2026-10-12/`
Historical procedure source	`achernar_data_collection.tex`

Decision

Use public human labels, suitable approved archive products and controlled simulation for broad perception development. Collect locally only the camera calibration, independent transfer examples and measured hardware outcomes that public data cannot provide. Reuse existing observations first. Expand a campaign only when an experiment identifies the missing evidence.

This plan replaces the original campaign quotas as prerequisites. It does not claim that any fixed smaller quota is sufficient for deployment. No training, camera calibration, hardware actuation or model promotion was performed by the source audit. Earlier documents remain dated evidence.

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1 Scope, priorities and standing decisions

The purpose is to minimize wasted acquisition while preserving honest scientific and operational evaluation. Begin with one actual camera/optical configuration and the modes intended for use. A supported one-rig prototype does not need a rented second rig; an unseen-rig claim does.

The original FQE 20/60/150-run tiers, COC thousands-per-class collection, TED 3,600-frame campaign, five-body calibration inventory and 10,000 local plate-solved-frame adapter trigger are no longer required starting conditions. They were planning envelopes, not empirical sample-size proofs. Their replacement is a sequence of small, decision-driven experiments.

1. Freeze the task, intended use, supported inputs and a baseline.
2. Inspect a small real payload and its rights/metadata before bulk intake.
3. Reproduce classical measurements and synthetic stress tests.
4. Run one public-trained candidate where it addresses a measured need.
5. Acquire/adjudicate only local development and independent evaluation gaps.
6. Check deployment parity, then observe in shadow mode before considering any additional control authority.

Native AstroM3 catalog research can proceed independently. FOP, learned exposure/filter/guiding policies and AstroM3Edge do not block useful cloud, focus-width, background or trailing measurements. No deadline or trained-model count is an acceptance criterion.

1.1 What remains authoritative

Code and migrations determine executable behavior. V6 defines proposed data contracts and decisions, not implemented ingestion or trained models. Classical HFR controls current autofocus; neural FQE is shadow output. BGE and TED retain classical baselines. Model lifecycle, integrity, runtime qualification, activation and controller authority remain separate.

No new hardware purchases, rentals, cloud forecasts or dedicated multimodal field campaigns are justified solely by this plan. A public source is admitted only for the task and intended use its evidence supports.

2 Verified starting point and remaining engineering work

Area	Current evidence and consequence
Local clouds	October 11 audit: 533 distinct readable R50 CR3s, 508 sky candidates, 200 provisional class candidates, 308 unknowns and 25 invalid ordinary-cloud examples. Labels are not approved truth; no reliable full-frame clear/thin set was established. Clock is unreliable and five gap groups are not five independent nights. Review before collecting more.
FQE	Actual 64-pixel grayscale shadow inference and a checksum/eligibility-aware SQLite-linked builder exist. Public FITS are not directly accepted; create an archive adapter without fake positions or focus-run records.

Area	Current evidence and consequence
Splitting	Temporal grouped splitting already exists. Enforce night/rig/ancestor constraints for the claim being evaluated; do not repeat the old absent-tool claim.
CFA analysis	RAW16 is read as grayscale; debayer module is a stub. Native CFA, validated color planes and display luminance are different measurement domains.
COC	Python model/trainer, metadata and Rust heuristic disagree on shape/task. Freeze a versioned input/label contract before training. No trained production cloud path is established.
TED / BGE	TED CNN entry delegates to classical moments. BGE uses robust global statistics; a SEP-style mesh background is not established by its wrapper name.
FOP / FDF	FOP dataset builder raises `NotImplementedError`. FDF is an implemented three-state CPU estimator, not a neural training task.

Source pointers and the limits of this targeted inspection are in the audit memos. Older capture/import defects must be rechecked before field use; the historical audit is not proof they all persist. In particular, existing import code is not evidence of correct CR3 metadata or wrong-clock label handling.

2.1 Implementation gates before new acquisition

1. Freeze analytics versus preview transforms, clipping units, source masks, crop geometry, missingness and label semantics. If R50 deployment uses JPEG preview, raw-only training does not establish preview-domain parity.
2. Add source manifests and task adapters. Preserve raw bytes, masks, calibration level, native units and original exposure identities.
3. Add reviewed cloud partial-label conversion and an explicit ignore mask. Do not silently fill unannotated areas or misspelled annotation classes.
4. Validate grouping/derivative ancestry before training. Freeze independent calibration/validation and final-test roles.
5. Confirm the field logger records actual settings, clock quality, position validity and disturbances. A schema field is not proof the hardware fills it.

3 Public-source intake: small, reversible and task-specific

3.1 First source choices

Source	Initial useful intake	Decision boundary
Lenghu human regions	22.1 MB ZIP, 1,111 embedded 512-pixel images and LabelMe regions. Inspect overlays and label corrections.	Primary COC pilot. Unknown pixels remain ignore; source split does not establish R50 transfer.
Lenghu bulk logits	None initially; later a stratified subset only if needed	Teacher predictions are weak supervision; exclude holdout ancestors and nights.
GalSim	Version-pinned analytic optical/atmospheric/detector scenes	Generator parameters are not measured real HFR or focuser steps.
MAST selected imaging	A named public mission/instrument/product with units, SCI/ERR/DQ where applicable	Prefer explicit public-domain product evidence; raw DN, electrons and electrons/s remain distinct.
MILAN / HSC / LCO	Bounded metadata/header research, then selected pixels only if useful and permitted	Separate noncommercial research lineage. No product eligibility by public-access inference.
AstroM3 native	One compatible full configuration/checkpoint; initially 100–1,000 objects for inference	Existing encoders; no local observations. Resolve identity/terms before a new image match.
MMU	At most one approved image subset and a small matched-object pilot	Optional after task utility/identifiability. Do not bulk-fetch the collection.
SWIN / Almeria	Optional external/night or daylight diagnostic	Shared SWIN ancestors stay out of training if SWINSEG is a test. Track separate terms.
DeepSpaceYolo / BTSbot / Trailblazer	Optional task-specific challenge examples	Object boxes/transients/satellites are not focus or tracking truth.
DeepSky / WOANC	No initial dependency	Daytime/unreleased or request-only; do not delay a useful experiment for extra sources.

3.2 Admission checklist and transfer caps

For imaging archives, start with metadata for 10–20 exposures from 3–5 sessions where available, and a proposed 2 GB total pilot transfer cap. The cap bounds effort, not training adequacy.

Choose a smaller individual-product route when one archive bundle exceeds it. Do not expand solely to fill a row counter. Full cloud logits, the year of MILAN, a multi-year Rubin database and 100 TB MMU are not initial requirements.

Before admitting each product, record: exact release/revision and URL; license for data/code/weights separately; checksum; instrument/site/session; HDU or array meaning; native units and scale; masks/variance; processing state; CFA/black/white/clipping evidence; timestamp reliability; duplicate ancestors; and available labels with their actual meaning. If clipping cannot be established, exclude it from strict FQE labels while optionally retaining an explicit robustness-only example.

A suitable image is not automatically a suitable outcome label. A processed variance map is an independent pipeline estimate, not measured sensor noise truth. A PSF model is a fitted source, not an independent exposure. A saturated or cloudy archive frame is useful for failure behavior only when its validity and label evidence are retained.

3.3 Rights and reproducibility

Lenghu has an Apache-2.0 dataset tag; teacher/backbone terms are separate. SWIN's original hosts link CC BY-NC 4.0. MILAN/DeepSpaceYolo are CC BY-NC-ND; HSC archive terms are noncommercial scientific/educational; LCO states CC BY-NC 2.0. AstroM3 source, processed-data listings and pretrained checkpoints have different terms; MMU preserves subset restrictions. A source-code license does not settle a downloaded database or user-log license.

Maintain a separately identifiable research lineage and an intended-use-cleared lineage. Retain permissions, citations and downstream source ancestry. This plan makes no blanket legal assertion about whether a particular weight is an adaptation. Additional permission may be recorded later without silently rewriting the provenance of an existing artifact.

4 Scientific labels and evaluation groups

4.1 Labels that must remain distinct

Label type	Meaning and limitation
Human cloud region	Conservative source annotation. Unannotated pixels are unknown. Cloud coverage is not transparency or observing loss.
Measured HFR	Named background/aperture/source-gate reference in pixels. Agreement with its labeler alone does not establish better focus.
Fitted PSF or uncertainty map	A named pipeline/model product, with its own measurement assumptions and calibration level.
Teacher logits	Versioned model output; confidence is not independently verified correctness or local probability calibration.
Latent simulation parameter	Defocus waves, motion path/length, true flux and noise parameters defined within the generator. Not motor steps or noisy moments.
Noisy estimated label	HFR/ellipticity from a named estimator after sampling, noise, clipping and masking; retain failed/invalid results.

Label type	Meaning and limitation
Focus optimum/outcome	Position-referenced converged run and verified final image/move. Neither one crop nor minimum sampled HFR alone supplies it.
Policy reward	Attributable measured result of an executed action under a versioned objective; unchosen-action rewards remain absent.

For FQE retain `log(HFR_px)`, uncertainty and accepted-source coverage. Position is provenance, not a pixels-only FQE input. Disable real within-run ranking on unrelated archive exposures. Maintain a separate simulated-sequence flag if ranking synthetic curves.

For TED retain latent trail length/axial angle, noiseless reference shape and noisy estimated ellipticity separately. Angle is modulo 180 degrees; almost round sources need an undefined/low-confidence angle, not arbitrary direction. Single-crop shape cannot identify tracking versus coma or wind. Use WCS before claiming sky-coordinate RA/Dec axes.

4.2 Split hierarchy and adaptive collection

All derivatives of a source exposure or latent scene family stay together; all modalities/repeated observations of an object stay together. Split before augmentation. FQE/FOP runs are nested within nights and rigs; use the coarser level needed for the claim. COC needs whole-night/weather-episode/camera-era separation. Similar sky frames separated by minutes are not independent conditions. Unknown dates require conservative group families.

Use development data for transforms and training; independent validation for selection/calibration; an untouched final set for a final claim. Never tune thresholds on the final test. If its failures guide new training, record that the test has been consumed and obtain fresh final evidence. Public benchmark labels already seen during development are not an independent local test.

Report source exclusions, usable frame coverage and independent group counts. A million crops are not a million observing outcomes. Confidence intervals must respect night/run/episode clustering. Fix the failure event and tolerance before choosing sample size. With zero failures in independent stationary trials, the one-sided 95% upper bound is $1 - 0.05^{1/n}$; 59 such trials are needed for a bound below 5%. This is an illustration, not a demand for 59 adjacent images or an assertion that 59 sessions settle every use.

5 Model-by-model acquisition decision

Model	First experiment / reuse	Additional local evidence only if needed
FQE	Classical benchmark, approved public stars, analytic PSF/noise scenes, one compact candidate	Supported-input bias/coverage and independently verified focus outcomes. Begin with a 6–10-sweep effort cap, not a release minimum.
FOP	Test canonical 38 features and synthetic/asymmetric/backlash curves against robust classical fitting	Readable position, sign/scale, reversals, validated optima and final outcomes. Missing position defers the model.

Model	First experiment / reuse	Additional local evidence only if needed
FDF	Existing $[p, v, k]$ estimator and simulated identification/failure tests	Harvest autofocus plus ambient-temperature records. Acquire extra thermal trajectories only if held-out forecasts or identifiability require them.
COC	Lenghu human regions and existing R50 adjudication	Missing fully clear/thin, moon/glow/contamination or independent held-out episodes. No class thousands before a transfer result.
TED	Classical moments with simulated motions/aberrations	One compact ambiguous-range untracked ladder plus round and confounder cases; expand only for residuals. Tracked-domain claims require real tracked evidence.
BGE	Classical full-frame benchmark and known-unit calibration products	Bench characterization of actual enabled mode, not every possible body/ISO. Neural noise map has no initial acquisition entitlement.
Exposure	Physics/rules baseline and passive action/outcome logs	Calibrated settings, safe support and known logging probabilities for randomized comparisons; no speculative exposure campaign.
FOR	Deterministic target/filter constraints and optional simulation	Only actual filter hardware/response/overhead and outcome logs. Keep deferred.
TME	Native encoders, approved configuration and original object identities	No dedicated field collection. Edge waits for identifiable stellar matches and image-only/catalog baselines.

Every cap in this table is an engineering proposal. Retaining the classical method after a negative model comparison is a valid decision. Collecting the cap does not confer shadow eligibility, promotion or motion authority.

6 Compact local runbooks

6.1 RB-A: adjudicate existing clouds before collecting more

1. Preserve original CR3s and existing provisional ledgers. Choose one reproducible RAW/preview transform, crop/sky mask and deployed resolution.
2. Review a diverse panel from all capture groups, including star-bearing openings, soft/low-signal images, foreground and bright/transient contamination. Compare native detail with the actual model input; do not label from a view the model cannot see without recording that difference.
3. Record cloud presence/region, visible-sky coverage if supportable, attenuation evidence, contamination and validity separately. Preserve legacy 0–4 candidates and alternates; leave

ambiguity unknown. No-stars alone is not overcast. Nearby weather is context, not a substitute for frame truth.
4. Adjudicate useful existing examples before new photographs. Keep uncertain clock/session groups conservative; never infer exact UTC from file mtime.
5. Freeze development and independent prospective-test roles before local adaptation. The current burst collection does not establish all five classes.

6.2 RB-B: targeted cloud transfer episodes

Only start after RB-A and a source-trained inference/input check name a gap. Fix camera clock/timezone, lens/pointing, manual focus, exposure recipe and RAW preservation. Log site, true time anchor, field/sky mask, camera settings, moon/illumination when computable, and observer evidence.

For a first tranche, cap effort at approximately 3–5 additional independent nights or weather episodes, not a compulsory campaign. Prioritize clear versus thin/star-attenuation transitions and one missing glow/obstruction contrast. A regular sparse cadence such as 30–60 seconds can preserve transitions; choose approximately 50–150 diverse keyframes per night after thinning if that many add information. These are workload bounds, not independence assumptions. Record short cloud transitions contemporaneously rather than spreading one 10-minute label across a changing field. Independent reference photos can help adjudicate but do not measure optical depth automatically.

Keep at least one untouched episode for each condition claimed and separate validation/calibration groups. If a condition appears in only one episode, report exploratory transfer rather than an independent performance estimate. Stop an uninformative uniform burst; further examples should answer a named error or uncertainty. Do not deliberately expose equipment to rain or damage to manufacture a challenge case; existing invalids and safe simulations suffice.

6.3 RB-C: supported-mode camera characterization

Start with one actual acquisition mode/ISO/gain and a locked decoder version. Record camera, lens/optics, firmware, compression/bit depth, sensor temperature when measured, exposure, CFA/black/white metadata and processing state. For a JPEG/video mode, characterize that transformed path separately; native RAW constants do not automatically describe it.

A proposed small bench starting budget is roughly 20 shortest dark/bias-like RAWs and 8–12 stable illumination levels with at least a pair at each level. Use paired differences to separate fixed spatial structure from temporal variance; record means after appropriate black correction and inspect clipping and nonlinearity. More repeats are justified if pair consistency or confidence is poor. Cameras without a true bias mode need a valid short-dark protocol; compressed SV105 video is not interchangeable with linear sensor RAW.

Characterize dark behavior only over exposure/temperature regimes intended for use; ordinary imaging darks can supply it passively. Do not build every ISO and exposure library before a task requires it. Flat-field calibration for science and PTC illumination measurements are related but distinct products. Preserve all originals and units. Document gain/read-noise fit assumptions and residuals; a single mean–variance slope is not proof of ideal Poisson behavior.

The gate is reproducible black/clipping/linearity evidence and a calibration bundle for the supported mode. Unknown values remain unknown; drop strict labels dependent on missing thresholds. Expand modes only when enabled in the product or a measured error points to the missing characterization.

6.4 RB-D: compact focus-width and outcome experiment

Confirm safe travel/focus reach and acquisition first; this plan does not assert the R50 reaches prime focus on the Z114. Use a configuration that actually produces resolved, unsaturated stars. Reuse any ordinary verified autofocus runs. If manual focus lacks readable position, record ordered settings for width research; do not claim signed-step FOP labels or autonomous verification.

Begin with a 6–10-run effort cap over approximately 2–3 distinct sessions when available. This can diagnose transfer, not qualify a general release. Use enough positions on both sides of an interior minimum, dense near focus; a starting 9–15 positions with 2–3 repeated frames can reveal scatter before expanding to the historical 21-by-5 procedure. Adjust sampling to actual curve resolution and repeatability, not a universal step count.

For each sample log capture/position/move times, actual readable position and validity, direction/reversal, settling, settings, temperature provenance, source/asset checksum and disturbances. Keep central and edge stars; retain rejection/coverage statistics. Record failed/aborted runs without fake optima. An interior fitted optimum needs uncertainty and independent final-image verification after the commanded move; then log the observed result.

Compare classical and neural estimates on matched accepted stars and full-frame coverage. Use held-out run/night groups for ranking, residuals and post-move outcome. No focus improvement means no expansion merely to train a larger net. Real control claims require reliable motor/sign/scale/backlash and actual final outcomes; external optical defocus in wavelengths cannot supply those steps.

6.5 RB-E: compact trailing/confounder experiment

Use the same useful local sessions where possible. Select central and edge stars; include short near-round exposures and denser steps where measured ellipticity approaches the existing ambiguous band. An initial one-session ladder across 2–3 fields and 6–10 exposure durations with a few repeats is a planning cap; record actual durations and source coverage.

An ideal untracked sidereal displacement is approximately $15.041 \cos \delta$ arcsec per second, converted through local image geometry. It does not exactly label observed ellipticity or identify wind versus tracking. Plate scale/WCS or another measured angular calibration is required for physical units; image-coordinate shape benchmarks can proceed without pretending those units are known. Alt-az tracking retains field rotation.

Retain coma, defocus, saturation, blends, weak stars and real wind/disturbance flags as confounders. A windy session is an optional gap-filling observation, not a compulsory night. Compare estimated shape/axial angle and frame false alerts with the classical baseline. Round/low-SNR angles can be undefined. No neural benefit on the measured ambiguous cases means stop the CNN branch.

6.6 RB-F: passive thermal and decision logging

During necessary autofocus, record verified best position, ambient temperature and probe/time quality, optical train/filter, resets and actual move success. The current FDF state is position, independent drift rate and thermal coefficient. A monotonic temperature/time trajectory can confound the last two; seek independent cooling/warming behavior only if identification or forecasts remain poor. Do not prescribe a fixed degrees-C span as universally sufficient. Hold out nights and compare against recent-best/no-drift baselines and the rig's measured critical-focus

threshold.

For exposure/filter/guiding decisions retain predecision context and measurement ages; safe/available actions; policy/version; requested/chosen/executed action; probabilities when randomized; actual settings/ownership/overrides; frame/event lineage; realized reward components and attribution window; and interruptions. Do not invent propensities for deterministic past choices. A fitted after-the-fact propensity model does not create randomization or remove hidden confounding. Passive fixed-policy data is useful for simulator checks without claiming unbiased new-policy evaluation. No bandit exploration starts merely to fill a training table.

7 Minimal proposed manifest and field-log contract

Store source/example manifests separately from operational action records. Task adapters may leave inapplicable fields absent; they must not replace missing measurements with plausible zeroes. Record at least:

Record	Required information
Source release	Provider, release/revision, direct URL, data/code/weight terms, intended-use disposition and evidence, original checksum.
Exposure	Source exposure ID, instrument/rig/mode, settings/units/calibration level, clock quality/time anchor, session/night family, masks and raw/derived links.
Derived example	Parent exposure/object/scene family, crop/resize geometry, transform/calibration version, simulation seed/parameters, derivative checksum.
Label	Type/method/version, value/units, uncertainty/validity, partial ignore mask or rejection, human/teacher provenance and correction ledger.
Partition	Dataset revision, ancestor group, development/validation/calibration/ final-test role, camera-era/object/night constraints and frozen assignment.
Focus outcome	Run/sample, read position/sign/scale, direction/reversal, fit/optimum uncertainty, executed move and independent final verification.
Decision	Predecision context, safe arms, policy/probabilities, requested and executed action, observed reward/overhead, interruption/override and parent assets.

A schema draft does not mean importers or SQL migrations support every field. Implement explicit adapters/validation and golden transform fixtures before creating training-ready datasets. Existing manually reviewed cloud CSVs are not automatically importer-compatible training input.

8 Stop, expand or proceed: acceptance gates

The matched comparisons and acquisition decision ledger are detailed in `docs/public-dataset-audit-2026-10-12/experiment-design.md`. For COC compare the current heuristic, public-human-only candidate and the same candidate with disjoint local adaptation on identical validation groups. For FQE compare classical measurement with one architecture under synthetic, approved public-real and mixed training. Do not enlarge architecture and data simultaneously and then attribute the result to a dataset. TED/BGE first test whether deterministic source/mask/background improvements resolve the error.

Every additional tranche must name a gap, explain why existing/public/synthetic evidence cannot answer it, specify the comparison it will decide, freeze its group/partition role, and state an effort cap and next decision. Data-efficiency curves use nested training groups and fixed validation groups. Do not repeatedly peek at a final test or stop a reliability test at its first favorable result.

Gate	Stop or resolve first	Evidence allowing the next stage
Intake	Unclear rights, units, labels, ancestors or clipping for strict labels	Small actual payload, reviewed metadata/terms and correct provenance.
Baseline	No reproducible classical measurement or reference semantics	Known-unit benchmark, failures/coverage and independent synthetic checks.
Candidate value	Only teacher agreement or simulator success; no operational need identified	Improved relevant error/coverage/latency tradeoff on frozen independent development/validation groups.
Local transfer	Missing clear/thin or mode parity; correlated-only test	Named gap filled and separate independent target-domain evidence.
Deployment	Unmatched preprocessing/export, uncalibrated uncertainty or unqualified runtime	Golden input/output parity, calibrated held-out behavior, resource qualification and inspectable shadow logs.
Control	Unknown motion, unbounded action, no attributable outcomes or support	Controller-specific real outcome evidence and existing safety/ownership gates; artifact selection alone never grants authority.
Expansion	Bigger corpus/rig list with no named decision benefit	Residual slice, missing supported condition or confidence interval justifies specific new data; freeze its role before collection.

For COC report local recall at a predeclared false-pause constraint, thin-cloud misses, contamination/unknown behavior and calibration; masks and whole-frame decisions need separate metrics. For FQE report accepted-cohort log-HFR error, uncertainty and full-frame coverage plus real focus outcomes. TED needs ellipticity/axial-angle errors and frame false alerts, including confounders. BGE needs calibrated unit/residual/gradient/crowding checks. FOP/FDF require

actual convergence/forecast outcomes; policy learners require defensible policy comparison. Native TME needs held-out-object retrieval/missing-modality tests, not only matching embedding dimensions.

9 Additional algorithms and deliberate deferrals

FW-1 needs calibrated per-filter response, not a new generic ML corpus. FW-2 is a policy with action/outcome requirements. FW-3 begins with explicit constraints/objectives and deterministic optimization scenarios. M-1/M-2 need actual guide/dither commands and attributable settle/science outcomes. M-3 needs genuine solve-failure/recovery need, not a low-star-count proxy. M-4 needs this mount's solved residuals over held-out sky coverage, coordinates and mount/pier configuration. Public logs and simulated fields help software tests; they do not supply the missing local labels or usage rights.

Do not start a separate campaign for any of these before supported hardware, truthful telemetry and a fixed baseline exist. Harvest during ordinary use. AstroM3Edge is deferred until object IDs, valid stellar matches, identifiable single-image information and an explicit alignment target exist. Neither a 10,000-frame quota nor a large galaxy archive answers those questions.

10 Sources, revision and boundaries

The complete source/inspection ledger, version IDs, hashes, license differences, file defects and unresolved questions are in:

- `docs/public-dataset-audit-2026-10-12/cloud-datasets.md`
- `docs/public-dataset-audit-2026-10-12/imaging-datasets.md`
- `docs/public-dataset-audit-2026-10-12/control-multimodal.md`
- `docs/public-dataset-audit-2026-10-12/README.md`
- `docs/public-dataset-audit-2026-10-12/experiment-design.md`

Principal primary sources: Lenghu paper/release https://arxiv.org/html/2603.16429v1, https://huggingface.co/datasets/ruiyicheng/LenghuSky-8; MILAN https://zenodo.org/records/6865830; HSC https://hsc-release.mtk.nao.ac.jp/doc/index.php/faq__pdr3/; LCO https://lco.global/observatory/termsofservice/; MAST https://stdatu.stsci.edu/publishing/data-use; GalSim https://github.com/GalSim-developers/GalSim; AstroM3 https://github.com/MeriDK/AstroM3; MMU https://github.com/MultimodalUniverse/MultimodalUniverse; NIST https://itl.nist.gov/div898/software/dataplot/refman2/auxillar/exacbici.htm.

The follow-up AI synthesis supplies useful three-workstream framing and shared/per-rig separation. Its asserted counts, fixed session suggestions and simplified lifecycle are not adopted as evidence. V6 preserves explicit scientific units and v5's orthogonal approval/qualification/authority states. The historical manual procedures remain available when additional sampling is justified; obsolete quotas or defects must not silently become current gates.

This is a plan and research result. Training, adapter implementation, camera calibration and hardware qualification are outstanding work. The first useful result is a small trustworthy experiment and a decision about what to retain, change or collect next.